Factor	Air Cooling Systems	Liquid Cooling Systems (AIO / Custom Loops)
Initial & Long-Term Cost	**Low to Moderate:** High-quality aftermarket air coolers or stock shroud designs are highly cost-effective, typically ranging from **₹2,500 to ₹8,500**. There are zero ongoing maintenance costs.	**High to Very High:** All-In-One (AIO) liquid coolers range from **₹8,500 to ₹20,000+**, while premium custom liquid loops can easily exceed **₹40,000**. Components like pumps may require complete replacement if they fail out of warranty.
Cooling Efficiency & Thermals	**Standard to Good:** Highly effective for standard gaming and mainstream workloads. However, under sustained maximum loads or heavy overclocking, temperatures can plateau higher as air has a lower specific heat capacity than liquid.	**Superior:** Liquid absorbs and transfers heat away from the GPU core much faster than air. It maintains significantly lower peak temperatures and takes much longer to reach thermal saturation, preventing thermal throttling during prolonged rendering or intense computing tasks.
Maintenance & Lifespan	**Minimal Upkeep / Long Lifespan:** Requires only periodic cleaning with compressed air to remove dust from the fans and aluminum fins. If a fan fails, it is cheap and simple to replace. The metal heatsink itself lasts indefinitely.	**Moderate to High Upkeep:** AIO systems are sealed but have a finite lifespan of roughly 4 to 6 years due to natural liquid permeation and mechanical pump wear. Custom loops require dedicated maintenance, including draining, flushing, and refilling the coolant every 12 to 24 months to avoid sediment buildup or galvanic corrosion.
Acoustic Performance (Noise)	**Variable:** Quiet at idle, but can become noticeably loud under intense workloads. The smaller fans must spin at high RPMs to move enough volume through dense metal fins to keep up with high thermal outputs.	**Consistently Quieter under Load:** Because radiators offer a much larger surface area to dissipate heat, the attached large-diameter fans can spin at much lower, quieter RPMs while delivering excellent thermal performance. *Note: A faint, consistent hum from the liquid pump is always present.*